A SECOND NINE-DIGIT FIXED POINT OF THE 1089-TRICK MAP, REFUTING A UNIQUENESS CONJECTURE IN OEIS A393794

MATH0

Abstract. Let $\mathrm{rev}(m)$ denote the decimal reversal of m (leading zeros dropped) and let $T(x) = D+\mathrm{rev}(D)$ with $D = |x-\mathrm{rev}(x)|$ be the 1089-trick map. In OEIS A393794, Havenaar conjectured that for every $n \geq 1$ the number $a(n) = 110\,(10^{n+1}-1)$ is the only fixed point of T with exactly $n+4$ decimal digits. We show that 108910890 is a nine-digit fixed point of T different from $a(5) = 109999890$, so the conjecture is false; the counterexample is ten times the term 10891089 of A001232, and an elementary argument, not formalized, shows that the same construction gives a second $(n+4)$-digit fixed point for every $n \geq 5$. To our knowledge this is the first explicit refutation, although the argument is elementary and the number had already been computed without being connected to the conjecture: it is listed among the fixed points of T in the conjecturer's own computational data, and, as one element of a 2-cycle of a closely related reversal map, in a 2024 paper of Almirantis and Li. The refutation is verified in Lean 4 against a statement that math0 formalized from the OEIS entry, checked against the OEIS data, reviewed for fidelity and frozen before the proof attempt.

1. Introduction

For $m \in \mathbb{N}$ let $\mathrm{rev}(m)$ be the number whose decimal digits are those of m in reverse order, with leading zeros of the result dropped (OEIS A004086 [11]); for example $\mathrm{rev}(10890) = 9801$. The *1089-trick map* is

$$T(x) = D + \mathrm{rev}(D), \qquad D = |x - \mathrm{rev}(x)|.$$

The name alludes to the classical puzzle in which a three-digit number whose first and last digits differ is replaced by the difference D of it and its reversal, written with three digits, and then D is added to its own reversal; the result is always 1089. The map T applies the same two operations to numbers of any length, but without padding.

In OEIS A393794 [8], Remco Havenaar (February 27, 2026) lists the numbers

$$a(n) = 110\,(10^{n+1}-1) \qquad (n \geq 1): \quad 10890,\ 109890,\ 1099890,\ 10999890,\ 109999890,\ \ldots,$$

whose decimal digits are $1, 0, 9^{\,n-1}, 8, 9, 0$. The entry proves that each $a(n)$ is a fixed point of T and conjectures that it is the *only* fixed point of T with $n+4$ digits, having verified this exhaustively for $n = 1, \ldots, 4$. The same family and the same conjecture appear in Havenaar's manuscript [4], in the repository linked from the entry [5]: writing $k = n+4$ for the number of digits, its Remark "Uniqueness per digit count" reads (with $n_k = a(k-4)$ and $\mathcal{D}_{10}^{k}$ the set of k-digit numbers) "Exhaustive computation for $k = 5, 6, 7, 8$ confirms that n_k is the *unique* fixed point of T in $\mathcal{D}_{10}^{k}$. We conjecture that this holds for all $k \geq 5$," and the question is repeated in the manuscript's list of open problems.

Both sources relate $a(n)$ to OEIS A001232 [9], the numbers $k > 0$ with $9k = \mathrm{rev}(k)$: 1089, 10989, 109989, 1099989, 10891089, ... They observe that $a(n)/10 = 11\,(10^{n+1}-1)$ is a *primitive* term of A001232, that is, one whose digits are $1, 0, 9^{\,n-1}, 8, 9$. The structure of A001232 is classical. D. W. Wilson (1997) showed that its terms are the numbers $99m$ in which m is a

Date: October 18, 2026.

palindrome over the digits $0, 1$ with no isolated digit, so that there are $F_{\lfloor j/2 \rfloor - 1}$ terms with $j \geq 4$ digits (F_i the Fibonacci numbers); M. F. Hasler (2022) described the terms as symmetric concatenations of primitive terms separated by blocks of zeros (both results are recorded in [9]). Sloane's survey [12] treats the more general question of which numbers are multiplied by a constant when their digits are reversed. Almirantis and Li [1, 2] study the iteration of the two-option reversal map $\Phi(y) = y - \mathrm{rev}(y)$ if $\mathrm{rev}(y) < y$, and $\Phi(y) = y + \mathrm{rev}(y)$ otherwise. This map is closely related to T: if $x > \mathrm{rev}(x)$ and $D = x - \mathrm{rev}(x)$ satisfies $\mathrm{rev}(D) \geq D$, then $\Phi(\Phi(x)) = T(x)$. In [2, Corollary 3.3] they show that if $\mathrm{rev}(y) = 9y$ then y and $10y$ form a 2-cycle of Φ, which is the computation in Lemma 3 below, and their Table 2 lists the 2-cycle $\{10891089, 108910890\}$. The variant of Φ that also subtracts when $\mathrm{rev}(y) = y$, and so sends palindromes to 0, is the map of OEIS A124074 [10], contributed by S. Dumortier in 2006, whose example already exhibits the 2-cycle $1089 \to 10890 \to 1089$. Section 2 of [2] also studies the two steps of the 1089 trick for numbers of any length, that is, the values $T(y)$ for $y > \mathrm{rev}(y)$ such that $y - \mathrm{rev}(y)$ has as many digits as y, and its Appendix A.2 lists 108910890 among these values for eight-digit y. Neither paper considers the fixed points of T, and [2] predates the conjecture of A393794. (Corollary, table and appendix numbers refer to the arXiv version of [2].)

Result and status. We show (Theorem 1) that $x_* = 108910890$ is a nine-digit fixed point of T with $x_* \neq a(5) = 109999890$, so the conjecture fails at $n = 5$. It is the smallest failure, since the conjecture holds for $n \leq 4$ by the exhaustive checks of [8], which the math0 referees repeated. The witness is $x_* = 10 \cdot 10891089$, ten times the smallest non-primitive term of A001232, and Lemma 3 explains why: ten times any term of A001232 is a fixed point of T. The same construction gives a second fixed point with $n + 4$ digits for every $n \geq 5$ (Proposition 5).

A393794 contains no correction or comment noting a counterexample: revision #19 (March 10, 2026) was still the current revision when we checked `https://oeis.org/A393794` in October 2026. The math0 referees found no refutation on arXiv or elsewhere on the web. A393794 is not part of the formal-conjectures repository [3]. We note, however, that the number 108910890 had already been computed. It is listed, without comment, among the fixed points of T in a results file of Havenaar's repository [5] (file `data/results_export.json`, experiment 10, time-stamped February 25, 2026, which iterates T from all $9 \cdot 10^6$ seven-digit starting values). It also appears twice in [2], which predates the conjecture: in Table 2, as the larger element of a 2-cycle of the map Φ described above, and in Appendix A.2, among the outputs of the two-step 1089 trick on eight-digit numbers. In none of these places is it connected with the conjecture. With these caveats we classify the result as new; mathematically it is elementary.

2. Statement

2.1. **The theorem.** For $x \in \mathbb{N}$ let $\ell(x)$ be the number of decimal digits of x, with $\ell(0) = 0$; thus $\ell(x) = L \geq 1$ if and only if $10^{L-1} \leq x < 10^L$. Let rev, T and $a(n)$ be as in the introduction, and put

$$x_* = 108910890.$$

Theorem 1. *$\ell(x_*) = 9$, $T(x_*) = x_*$, and $x_* \neq a(5) = 109999890$.*

Corollary 2. *The following statement is false: for every $n \geq 1$, $\ell(a(n)) = n+4$, $T(a(n)) = a(n)$, and every $x \in \mathbb{N}$ with $\ell(x) = n + 4$ and $T(x) = x$ equals $a(n)$.*

The first two clauses of Corollary 2 hold for every $n \geq 1$ (Lemma 4), so it is the uniqueness clause that fails.

2.2. **The OEIS conjecture.** The conjecture is the first comment of A393794 [8], an entry by Remco Havenaar dated February 27, 2026 (revision #19, March 10, 2026). The entry is named "Fixed points of the 1089-trick map T(n) = |n - reverse(n)| + reverse(|n - reverse(n)|)", has offset 1 and formula "`a(n) = 110*(10^(n+1) - 1)`", and the comment reads verbatim:

> a(n) is the unique fixed point of the 1089-trick map T among all (n+4)-digit numbers (verified exhaustively for n = 1..4; conjectured for all n >= 1).

The entry's example "10890 - 09801 = 1089; 1089 + 9801 = 10890" shows that reversal drops leading zeros: with rev(1089) padded to five digits, 98010, the example would fail.

2.3. **The Lean statement.** The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A393794.lean` (namespace `Math0.Statements.A393794`). It was reviewed for fidelity and frozen (SHA-256 `3c0423ef...ae69a`) before any proof attempt. Its definitions and the statement are (verbatim):

```
/-- `reverse(m)` (A004086): the number whose decimal digits are those of `m` in reverse order
(leading zeros of the result are dropped). -/
def rev (m : ℕ) : ℕ := Nat.ofDigits 10 (Nat.digits 10 m).reverse

/-- The 1089-trick map `T(x) = |x - reverse(x)| + reverse(|x - reverse(x)|)`. -/
def T (x : ℕ) : ℕ := Nat.dist x (rev x) + rev (Nat.dist x (rev x))

/-- A393794: `a(n) = 110 * (10^(n+1) - 1)` (`n ≥ 1`), the fixed points `10890, 109890,
1099890, ...` of the 1089-trick map `T` listed in the entry; `a(n)` has `n + 4` digits. -/
def a (n : ℕ) : ℕ := 110 * (10 ^ (n + 1) - 1)

/-- Conjecture (Havenaar, A393794): for every `n ≥ 1`, `a(n)` is the unique fixed point of the
1089-trick map `T` among all `(n + 4)`-digit numbers. -/
def statement : Prop :=
  ∀ n : ℕ, 1 ≤ n →
    (Nat.digits 10 (a n)).length = n + 4 ∧ T (a n) = a n ∧
      ∀ x : ℕ, (Nat.digits 10 x).length = n + 4 → T x = x → x = a n
```

2.4. **How the encoding matches the OEIS.** *Reversal.* Mathlib's `Nat.digits 10 m` is the list of decimal digits of m, least significant first, with no trailing zeros (so `Nat.digits 10 0` is empty), and `Nat.ofDigits 10 L` evaluates $\sum_i L_i 10^i$. Reversing the list and evaluating it gives the digits of m read backwards, and any zeros that become leading zeros contribute nothing. So `rev` is A004086, and the kernel confirms `rev 10890 = 9801` in agreement with the entry's example. *The map.* `Nat.dist x y` is $|x - y|$, so truncated subtraction on $\mathbb{N}$ plays no role in `T`. *The sequence.* `a` is the entry's formula with offset 1. As $10^{n+1} \geq 1$, the subtraction does not truncate. The entry's data, its Mathematica and Python programs, and its proof that $T(a(n)) = a(n)$ all use this formula. The name, read as "$a(n)$ is the n-th positive fixed point of T", gives an equivalent conjecture: under the conjecture, the positive fixed points with at least five digits are exactly the $a(n)$, and `least_fixed_point` below shows there is none with fewer digits; conversely, under the other reading the only $(n+4)$-digit fixed point must be $a(n)$, which is one by Lemma 4. The formula reading is used because defining $a(n)$ as the n-th fixed point would build the conjectured uniqueness into the definition. *Digit length.* "$(n+4)$-digit" is `(Nat.digits 10 x).length = n + 4`, i.e. $\ell(x) = n + 4$. This excludes the trivial fixed point $T(0) = 0$. *Range.* The comment says "conjectured for all n >= 1", and the statement adds no other hypotheses. The two conjuncts asserting that $a(n)$ has $n + 4$ digits and is fixed are part of the meaning of "$a(n)$ is the unique fixed point"; they are true (Lemma 4). *Data.* The statement file proves, each by `decide +kernel`: that $a(1), \ldots, a(18)$ are the 18 listed terms $10890, \ldots, 109999999999999999890$ (`a_data`); that T fixes each of them (`T_data`); that $a(i+1)$

has $i+5$ digits for $i < 18$ (`digits_data`); and that $10890 = a(1)$ is the least positive fixed point of T, by checking every $x < 10890$ (`least_fixed_point`).

Corollary 2 is the Lean theorem `Math0.Frontier.A393794.result`, whose type is `¬ statement`. Theorem 1 is the private lemma `witness` in the same file (Section 4). Lemmas 3 and 4 and Proposition 5 below are not formalized.

3. Proof

3.1. **The witness.**

Proof of Theorem 1. Since $10^8 \le x_* < 10^9$, we have $\ell(x_*) = 9$. The decimal digits of x_* are $1, 0, 8, 9, 1, 0, 8, 9, 0$; read backwards they are $0, 9, 8, 0, 1, 9, 8, 0, 1$, and dropping the leading zero gives

$$\operatorname{rev}(x_*) = 98019801.$$

As $x_* > \operatorname{rev}(x_*)$,

$$D = |x_* - \operatorname{rev}(x_*)| = 108910890 - 98019801 = 10891089.$$

The digits of D are $1, 0, 8, 9, 1, 0, 8, 9$, so $\operatorname{rev}(D) = 98019801$ (here no zero is dropped), and

$$T(x_*) = D + \operatorname{rev}(D) = 10891089 + 98019801 = 108910890 = x_*.$$

Finally $a(5) = 110\,(10^6 - 1) = 110 \cdot 999999 = 109999890 \neq x_*$. □

In Lean, the three facts of Theorem 1 form a single proposition, the lemma `witness`, which is closed by `decide +kernel`. Its second conjunct, `T 108910890 = 108910890`, is decided by evaluating the definitions: the kernel computes `Nat.digits 10 108910890` $= [0, 9, 8, 0, 1, 9, 8, 0, 1]$, reverses it to $[1, 0, 8, 9, 1, 0, 8, 9, 0]$, and evaluates $\sum_i L_i 10^i = 98019801$, exactly as in the proof above. The other two conjuncts are evaluated the same way.

Proof of Corollary 2. Suppose the statement holds. Apply it with $n = 5$, which satisfies $n \ge 1$, and take the uniqueness clause with $x = x_*$. By Theorem 1, $\ell(x_*) = 9 = 5 + 4$ and $T(x_*) = x_*$, so $x_* = a(5)$. This contradicts $x_* \neq a(5)$. □

The Lean proof of `result` is exactly this: it introduces the hypothesis `h : statement`, destructures `witness` into `hlen`, `hT` and `hne`, discharges $1 \le 5$ by `norm_num`, and applies `hne` to the third component of `h 5`, instantiated at $x = 108910890$.

3.2. **Where the counterexample comes from.**

The rest of this section is not part of the Lean development. It explains the witness and shows that the conjecture fails for every $n \ge 5$.

Lemma 3. *If $k \ge 1$ and $9k = \operatorname{rev}(k)$, then $T(10k) = 10k$ and $\ell(10k) = \ell(k) + 1$.*

Proof. If the decimal string of k is s, then that of $10k$ is $s0$. Its reversal is 0 followed by the reversal of s, which, once leading zeros are dropped, is the number $\operatorname{rev}(k)$. Hence $\operatorname{rev}(10k) = \operatorname{rev}(k) = 9k$, and $D = |10k - 9k| = k$. Then $\operatorname{rev}(D) = 9k$ and $T(10k) = k + 9k = 10k$. The digit count is clear. □

The same computation appears in [2, Corollary 3.3], phrased as the statement that k and $10k$ form a 2-cycle of the map Φ of the introduction.

Lemma 4 (A393794)**.** *For every $n \ge 1$, $\ell(a(n)) = n + 4$ and $T(a(n)) = a(n)$.*

Proof. Let $R = 10^{n+1} - 1$ and $k = 11R = 10^{n+2} + (10^{n+1} - 11)$. As $10^{n+1} - 11$ has the $n+1$ digits $9^{\,n-1}, 8, 9$, the digits of k are $1, 0, 9^{\,n-1}, 8, 9$. Similarly $9k = 99R = 98 \cdot 10^{n+1} + (10^{n+1} - 99)$, and $10^{n+1} - 99$, written with $n + 1$ digits, is $9^{\,n-1}, 0, 1$; so $9k$ has digits $9, 8, 9^{\,n-1}, 0, 1$, which are those of k reversed, and $9k = \operatorname{rev}(k)$. Lemma 3 applies to k, and $10k = 110R = a(n)$ has $\ell(k) + 1 = n + 4$ digits. □

This is the proof given in the entry [8], rearranged. The witness x_* arises in the same way from $k = 10891089 = 1089 \cdot 10001$, which satisfies $9k = 98019801 = \mathrm{rev}(k)$.

Proposition 5. *For every $n \geq 5$, the number $x_n = 10890\,(10^{n-1}+1)$ has $n+4$ digits, satisfies $T(x_n) = x_n$, and is different from $a(n)$. In particular $x_5 = x_*$.*

Proof. Put $m = n-1 \geq 4$ and $k = 1089\,(10^m+1) = 1089 \cdot 10^m + 1089$. As $1089 < 10^4 \leq 10^m$, the decimal string of k is $1089\,0^{m-4}\,1089$, with $m+4$ digits. Its reversal is $9801\,0^{m-4}\,9801$, that is, $\mathrm{rev}(k) = 9801 \cdot 10^m + 9801 = 9k$. By Lemma 3, $x_n = 10k$ is a fixed point of T with $m+5 = n+4$ digits. Finally

$$a(n) - x_n = (11000 \cdot 10^{n-1} - 110) - (10890 \cdot 10^{n-1} + 10890) = 110\,(10^{n-1} - 100) > 0.$$

□

So the uniqueness claim, which holds for $n \leq 4$ by exhaustive search, fails for every $n \geq 5$.

Remark 6. By Lemma 3 and Wilson's description of A001232 quoted in the introduction, the number of fixed points of T with $n+4$ digits is at least $F_{\lfloor (n+3)/2 \rfloor - 1}$, which is 2 for $n = 5, 6$ and tends to infinity with n. Exhaustive searches run by the math0 referees (one in C over all $x < 10^{10}$) found exactly the positive fixed points

10890, 109890, 1099890, 10999890, 108910890, 109999890, 1089010890, 1099999890,

which are ten times the terms of A001232 below 10^9. We do not know whether every positive fixed point of T is of this form. These searches also show that the data of A393794, read as the list of all fixed points of T suggested by the entry's name, omit 108910890 between $a(4)$ and $a(5)$.

4. Machine verification

Files and declarations. In the math0 repository (`https://github.com/Foundation-0/math0`), the proof is in `lean/Math0/Frontier/OEIS/A393794.lean`, namespace `Math0.Frontier.A393794`. The main theorem is `Math0.Frontier.A393794.result`, with the private lemma `witness`. The statement is `Math0.Statements.A393794.statement` in `lean/Math0/Statements/OEIS/A393794.lean`. The proof file, after its imports and module docstring, is (verbatim):

```
namespace Math0.Frontier.A393794
open Math0.Statements.A393794

/-- `108910890` has 9 digits, is fixed by `T`, and differs from `a 5` (kernel-checked). -/
private theorem witness :
    (Nat.digits 10 108910890).length = 5 + 4 ∧ T 108910890 = 108910890 ∧
      108910890 ≠ a 5 := by
  decide +kernel

/-- The conjecture fails at `n = 5`: `108910890` is a second 9-digit fixed point of `T`. -/
theorem result : ¬ statement := by
  intro h
  obtain ⟨hlen, hT, hne⟩ := witness
  exact hne ((h 5 (by norm_num)).2.2 108910890 hlen hT)

end Math0.Frontier.A393794

#math0_settles Math0.Frontier.A393794.result Math0.Statements.A393794.statement
```

Kernel computation. The tactic `decide +kernel` asks the Lean kernel to reduce the `Decidable` instance of the proposition in `witness` to `true`. The kernel unfolds the definitions of `Nat.digits`, `List.reverse`, `Nat.ofDigits`, `Nat.dist`, `rev`, `T` and `a`, using its built-in arbitrary-precision arithmetic on natural-number literals for the divisions, remainders, sums, products and comparisons. No compiled code is trusted: the file uses no `native_decide` (which would add the axiom `Lean.ofReduceBool`), and it contains no `sorry`, `axiom` or `set_option`. The same mechanism proves the data checks `a_data`, `T_data`, `digits_data` and `least_fixed_point` in the statement file.

Audit. The command `#math0_settles`, defined in `lean/Math0/Audit.lean` for statements that math0 formalizes itself, fails unless (1) `statement` is a universe-monomorphic definition of type `Prop`; (2) `statement` depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`, so it cannot hide a `sorry`; (3) the type of `result` is, up to metadata annotations and with no unfolding, either `statement` or ¬ `statement`; and (4) `result` also depends only on those three axioms. On this file it reports:

```
math0 audit: 'Math0.Statements.A393794.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A393794.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A393794.result' refutes 'Math0.Statements.A393794.statement'
```

Toolchain and reproduction. We use Lean 4 [7] `v4.33.1` and Mathlib [6] `v4.33.1`. The repository also pins formal-conjectures [3] at commit `c77fdd4`, which this result does not use. To reproduce, run `scripts/verify.sh` from the repository root. It builds all math0 results with `lake build`, which re-runs every audit command, and checks that the pinned dependencies are unmodified. To check this file alone once its imports are built (for example by `lake build`), run `lake env lean Math0/Frontier/OEIS/A393794.lean` in `lean/`; this used under a minute of CPU time on our machine. The math0 referees also recomputed $\mathrm{rev}(x_*)$, D, $\mathrm{rev}(D)$, $T(x_*)$ and $a(5)$ independently in Python.

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Produced by Claude (an AI model developed by Anthropic) in the math0 project